Conics and the Dedekind Construction of Addition

N-20220715

§1.1 Conics as metric loci and their tangent geometry

The metric definitions and coordinates of conics below follow OpenStax, *Calculus Volume 3*, Section 1.5; the real-number material later in this chapter follows MIT 18.100A.[1] Conic geometry and the construction of real numbers use coordinates in different ways: conics study metric loci, while Dedekind cuts construct the ordered scalar field in which those coordinates live. A cone cut by a plane produces the classical conics:

$$\text{circle}, \quad \text{ellipse}, \quad \text{parabola}, \quad \text{hyperbola}.$$

An ellipse can be defined as the locus of points P for which the sum of distances to two foci is constant:

$$PF_1 + PF_2 = 2a.$$

With foci $F_1 = (-c, 0)$, $F_2 = (c, 0)$, this gives the standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad c^2 = a^2 - b^2.$$

The eccentricity is

$$e = \frac{c}{a}, \qquad 0 < e < 1.$$

A hyperbola is defined by a constant difference of distances to the two foci:

$$|PF_1 - PF_2| = 2a,$$

with standard equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

or

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

depending on the transverse axis.

A parabola is the locus of points equidistant from a focus and a directrix. Its standard equations are

$$y^2 = 2px, \qquad y^2 = -2px, \qquad x^2 = 2py, \qquad x^2 = -2py.$$

For $y^2 = 2px$, the focus is $(p/2, 0)$, the directrix is $x = -p/2$, and the eccentricity is 1.

For an ellipse, a vertical line through a focus intersects the ellipse in endpoints of a latus rectum. The length is

$$\frac{2b^2}{a}.$$

[1] OpenStax, *Calculus*, Vol. 3, §1.5 (conics); MIT 18.100A real-analysis notes.

To derive it, substitute the focal abscissa $x = c$ into $x^2/a^2 + y^2/b^2 = 1$. Since $c^2 = a^2 - b^2$, this gives $y^2 = b^4/a^2$. The two endpoints $(c, \pm b^2/a)$ are separated by the claimed length. The tangent at $P = (x_0, y_0)$ to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1.$$

The note also records the focus-directrix relation. For an ellipse with eccentricity $e = c/a$, the directrices are

$$x = \pm \frac{a^2}{c}.$$

The ratio

$$\frac{PF}{\operatorname{dist}(P, \text{directrix})} = e$$

characterizes a nondegenerate conic with a chosen focus and its corresponding directrix. For the parabola $y^2 = 2px$ with $p > 0$, choose $F = (p/2, 0)$ and directrix $x = -p/2$. Equality of squared distances gives

$$(x - p/2)^2 + y^2 = (x + p/2)^2,$$

which simplifies to $y^2 = 2px$. This makes the metric-to-algebra transformation explicit.

A conic can be defined by the ratio

$$e = \frac{\text{distance to focus}}{\text{distance to directrix}}.$$

If $e < 1$, the conic is an ellipse. If $e = 1$, it is a parabola. If $e > 1$, it is a hyperbola. This unifies the three classical curves by one parameter: eccentricity.

§1.2 Affine Coordinates for Conics

A conic can be represented by a quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0.$$

Affine transformations preserve conic type up to degeneracy, while projective transformations unify ellipses, parabolas, and hyperbolas as sections of a cone.

The elementary focus-directrix and latus-rectum formulas are metric shadows of a more invariant projective object.

§1.3 Rational density and the real coordinate line

The analytic side begins with the density of rational and irrational numbers in $\mathbb{R}$. For rationals, given $x \in \mathbb{R}$ and $\varepsilon > 0$, choose $q \in \mathbb{N}$ with

$$\frac{1}{q} < \varepsilon.$$

Then choose an integer p such that

$$p \le qx < p + 1.$$

Thus

$$\left|x - \frac{p}{q}\right| < \frac{1}{q} < \varepsilon.$$

For irrationals, take a rational approximation r to $x - \sqrt{2}/N$, then

$$r + \frac{\sqrt{2}}{N}$$

is irrational and can be made arbitrarily close to x.

The density of $\mathbb{Q}$ in $\mathbb{R}$ says every real interval contains rationals. Dedekind cuts use order rather than decimal expansions to construct real numbers. Addition of cuts must be defined so that order and completeness are preserved.

This is the structural reason the note can move from conics to real-number construction: analytic geometry needs a complete ordered field.

The note includes the vector space axioms. Let F be a field and V a set equipped with addition and scalar multiplication:

$$+ : V \times V \to V, \qquad \cdot : F \times V \to V.$$

The axioms are associativity and commutativity of addition, existence of 0, additive inverses, compatibility of scalar multiplication, distributivity, and $1v = v$. With the usual coordinate operations,

$$\mathbb{R}^n$$

is an $\mathbb{R}$-linear space.

§1.4 Dedekind cuts: defining addition and additive inverses

The note returns to Dedekind cuts. We denote by $\mathbb{R}$ the set of Dedekind cuts. For cuts X, Y, define an order by

$$X < Y \quad \Longleftrightarrow \quad X \subset Y, \quad X \neq Y.$$

This order is total because if $X \not\subseteq Y$, choose $x \in X \setminus Y$; the downward-closed property of cuts forces $Y \subset X$.

Define

$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$

The zero cut is

$$\overline{0} = \{x \in \mathbb{Q} : x < 0\}.$$

One must check that $X + Y$ is again a Dedekind cut. It is nonempty; it is proper because rational upper bounds u_X, u_Y give an upper bound $u_X + u_Y$; it is downward closed because $q < x + y$ can be written as $(q - y) + y$ with $q - y < x \in X$; and it has no greatest element because X has none. The cut axioms below are all used:

(i) it is nonempty and not all of $\mathbb{Q}$;

(ii) it is downward closed;

(iii) it has no greatest element.

The page gives the standard proof. For downward closure, if $z < x+y$, then $z = x+(z-x)$ and $z - x < y$, so $z - x \in Y$. For no greatest element, choose larger elements $x' > x$ in X and $y' > y$ in Y; then $x' + y' > x + y$.

Associativity and commutativity follow directly:

$$(X + Y) + Z = X + (Y + Z), \qquad X + Y = Y + X.$$

Order compatibility is subtler. One needs the lemma: if X is a cut and $n \in \mathbb{N}$, then there exist $x \in X$ and $x' \in X'$ such that

$$0 < x' - x < \frac{1}{n}.$$

This approximation lemma lets one prove

$$X < Y \quad \Rightarrow \quad X + Z < Y + Z.$$

For a cut X, define

$$-X = \{y - x' : y \in \overline{0},\ x' \in X'\}.$$

Then $-X$ is the unique cut Z such that

$$X + Z = \overline{0}.$$

The sign construction uses rational approximation at the cut boundary: for a rational cut $X_a = \{q : q < a\}$, the set $\{y - x' : y < 0,\ x' \geq a\}$ is exactly $\{q : q < -a\}$, since one may choose y arbitrarily close to 0 and x' arbitrarily close to a from above. This reproduces the usual negative of a real number. In particular, if $p/q \in \mathbb{Q}$, then

$$-X_{p/q} = X_{-p/q}.$$

If A and B are lower cuts, their sum is

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

The definition is not arbitrary. It is forced by wanting the embedding of rationals into cuts to preserve addition and order.